

Equilibrium Distribution for t-Distributed Stochastic Neighbor Embedding with Generalized Kernels

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Abstract

T-distributed stochastic neighbor embedding (t-SNE) is a well-known algorithm for visualizing high-dimensional data by finding low-dimensional representations (van der Maaten and Hinton [2008]). In this paper, we study the convergence of t-SNE with generalized kernels and extend the results of Auffinger and Fletcher [2023]. Our work starts by giving a concrete formulation of generalized input and output kernels. Then we prove that under certain conditions, the t-SNE algorithm converges to an equilibrium distribution for a wide range of input and output kernels as the number of data points diverges.

1 The Concept of Input and Output Kernels

T-distributed stochastic neighbor embedding (t-SNE) is a powerful technique for visualizing high-dimensional data in a low-dimensional space, developed in 2008 by Laurens van der Maaten and Geoffrey Hinton (van der Maaten and Hinton [2008]). Despite its wide application (in biology for example Linderman et al. [2019], Wallach and Lilien [2009]), the theoretical aspects and mathematical rigor of t-SNE have rarely been discussed thoroughly. Recently, Auffinger and Fletcher proved that if data is sampled independently from a compact probability measure μ_X , then the output of t-SNE converges to an equilibrium measure that is also compact as the number of data points diverges (Auffinger and Fletcher [2023]). This, together with Arora et al. [2018] and Linderman and Steinerberger [2017], are some of the very first results that rigorously establishes the theoretical guarantees of t-SNE.

We aim to generalize the results from Auffinger and Fletcher [2023]. In the original version of the t-SNE algorithm, for each pair of input or output data points, a weight is assigned based on the distance between them. This weight is used to construct probability distributions that will be used in the loss function. A natural question to ask is: can we generalize the result for different weights such that the convergence still holds and the equilibrium measure still exists? A generalized formulation allows us to further reveal the theoretical nature of this algorithm, greatly enhancing our understanding of t-SNE and dimension reduction. Our work starts by generalizing the algorithm with the concept of input and output kernels.

Let $d, s \geq 2$ with $d > s$, and $n \geq 1$. Let $X_1, \dots, X_n \in \mathbb{R}^d$ be the input data points and $Y_1, \dots, Y_n \in \mathbb{R}^s$ the output data points. We define the input and output kernels as follows:

Definition 1 (Generalized Input Kernel). *The input kernel $\mathbf{Ker}_I(x, y, \sigma) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^+$ is a function of the input data points and parameter σ such that for any $x, y \in \mathbb{R}^d$ and $\sigma > 0$,*

$$\mathbf{Ker}_I(x, y, \sigma) = \mathbf{Ker}_I(y, x, \sigma). \quad (1.1)$$

Definition 2 (Generalized Output Kernel). *The output kernel $\mathbf{Ker}_O(z, w) : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}^+$ is a function of the output data points such that for any $z, w \in \mathbb{R}^s$,*

$$\mathbf{Ker}_O(z, w) = \mathbf{Ker}_O(w, z). \quad (1.2)$$

Clearly, by letting

$$\mathbf{Ker}_I(x, y, \sigma) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right), \quad \mathbf{Ker}_O(z, w) = \frac{1}{1 + \|z - w\|^2}, \quad (1.3)$$

we recover the original t-SNE algorithm by using both $\mathbf{Ker}_I$ and $\mathbf{Ker}_O$ as the weights for the input and output data points. We can also say that the original t-SNE algorithm has a Gaussian input kernel and a t-distributed output kernel.

From now on, we will use the terminology *generalized t-SNE* when we use the generalized formulation for input and output kernels to distinguish it from the original algorithm.

Clearly, not all choices of $\mathbf{Ker}_I$ and $\mathbf{Ker}_O$ will yield a valid generalized t-SNE algorithm, such as $\mathbf{Ker}_I$ that diverges as $\|x - y\| \rightarrow \infty$. To prove similar results as in [Auffinger and Fletcher \[2023\]](#), we need to impose some conditions on both $\mathbf{Ker}_I$ and $\mathbf{Ker}_O$. The conditions should provide just enough constraints for the convergence to hold, while applying to a wide range of function choices. Towards this end, we use the following conditions for the rest of our work. We will provide a detailed discussion on the generality of these conditions in [section 3](#).

Condition 1 (Input Kernel Conditions). *Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be a fixed function. The input kernel has the form of*

$$\mathbf{Ker}_I(x, x', \sigma) = \exp(-w(\sigma \|x - x'\|^\theta)) \quad (1.4)$$

Here, $\theta > 0$ is a fixed constant. w satisfies:

1. $\forall t \geq 0, w'(t) > 0. \lim_{t \rightarrow \infty} w(t) = \infty$.
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0$.
3. The following integral converges:

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (1.5)$$

Since the all probability masses are normalized, we can assume without loss of generality that $w(0) = 0$ from now on.

Condition 2 (Output Kernel Conditions). *The output kernel depends only on the distance between two output data points y and y' . In particular, $\mathbf{Ker}_O(y, y') = k(\|y - y'\|)$ and:*

1. $k : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. $\mathbf{Ker}_O$ satisfies

$$\iint_{(0, \infty)^2} \mathbf{Ker}_O(y, y') dy dy' < \infty. \quad (1.6)$$

3. k has bounded derivative.
4. $k'(0) = 0$.

Setup and Notations

In this subsection, we provide an introduction to the generalized t-SNE algorithm incorporating the new terminology. Recall that the input of the generalized t-SNE consists of vectors $X_1, \dots, X_n \in \mathbb{R}^d$, with the purpose of finding vectors $Y_1, \dots, Y_n \in \mathbb{R}^s$ that serve as a good representation of the original vectors. The algorithm starts by defining the following probability masses:

$$p_{j|i} = \frac{\mathbf{Ker_I}(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker_I}(X_i, X_k, \sigma_i)}. \quad (1.7)$$

Each σ_i satisfies that for each $i = 1, \dots, n$,

$$- \sum_{j=1, j \neq i}^n p_{j|i} \log p_{j|i} = \log \mathbf{Perp} =: \log(n\rho). \quad (1.8)$$

$\mathbf{Perp}$ is called the perplexity and $\rho \in (0, 1)$ is a given constant. The purpose of σ_i is to make sure that for each i , the probability distribution induced by $p_{j|i}$ has the same entropy across all i . Clearly, $p_{j|i}$ is not symmetric.

Remark 1. *For finite data set perplexity is usually set as a constant. Moreover, the scaling of perplexity with respect to $\log n$ is strictly necessary here. In Murray and Pickarski [2024], it has been proved that if we let $\mathbf{Perp} = \log n^\gamma \rho$ for some constant parameter γ , we cannot prove convergence and the existence of equilibrium measures for neither $\gamma < 1$ nor $\gamma > 1$.*

We then define $p_{ii} = 0$ and for $1 \leq i \neq j \leq n$,

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n} = \frac{1}{2n} \left(\frac{\mathbf{Ker_I}(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker_I}(X_i, X_k, \sigma_i)} + \frac{\mathbf{Ker_I}(X_i, X_j, \sigma_j)}{\sum_{k \neq j} \mathbf{Ker_I}(X_j, X_k, \sigma_j)} \right). \quad (1.9)$$

Obviously,

$$\sum_{\substack{j=1 \\ j \neq i}}^n p_{j|i} = 1, \quad \sum_{1 \leq i \neq j \leq n} p_{ij} = 1. \quad (1.10)$$

As for the output vectors $Y_1, \dots, Y_n \in \mathbb{R}^s$, set for $1 \leq i \neq j \leq n$,

$$q_{ij} = \frac{\mathbf{Ker_O}(Y_i, Y_j)}{\sum_{k \neq l} \mathbf{Ker_O}(Y_k, Y_l)}. \quad (1.11)$$

The goal of the generalized t-SNE is to find Y that minimizes the relative entropy

$$L_{n,\rho}(X, Y) = \sum_{1 \leq i \neq j \leq n} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right). \quad (1.12)$$

More precisely, the output of t-SNE Y^* is defined as

$$Y^* = \arg \min_{Y \in (\mathbb{R}^s)^n} L_n(X, Y). \quad (1.13)$$

In the expression of $L_n(X, Y)$, all p_{ij} are given and q_{ij} is a function of Y_i and Y_j . Therefore, Y^* can be found simply by using gradient descent. This completes the description of the algorithm itself. Now we introduce necessary notations for convergence and asymptotic analysis. Let $\mathcal{M}(\mathbb{R}^d)$ be the space of Borel probability measures on $\mathbb{R}^d$ and $\mu_X \in \mathcal{M}(\mathbb{R}^d)$ be a probability measure. Suppose all input $X = (X_1, \dots, X_n)$ are i.i.d. random variables

with probability measure μ_X . Meanwhile, let $Y^* = (Y_1, \dots, Y_n)$ be the output of the t-SNE algorithm. Consider the following empirical measure:

$$\mu_n := \frac{1}{n} \sum_{i=1}^n \delta(X_i, Y_i) \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s). \quad (1.14)$$

We are interested in its limiting measure as $n \rightarrow \infty$. To finish the setup, there are a few more remaining quantities we would like to introduce. Let $\mu \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s)$. For a given kernel $\mathbf{Ker}_I(x, y)$, define $F_{\rho, \mu}(x, \sigma) : \mathbb{R}^d \times \mathbb{R}^+ \rightarrow \mathbb{R}$ as

$$F_{\rho, \mu}(x, \sigma) = \int \frac{\mathbf{Ker}_I(x, x', \sigma)}{\int \mathbf{Ker}_I(x, x', \sigma) d\mu(x')} \log \left(\frac{\mathbf{Ker}_I(x, x', \sigma)}{\int \mathbf{Ker}_I(x, x', \sigma) d\mu(x')} \right) d\mu(x') + \log \rho. \quad (1.15)$$

Note that the last s variables in the integrands stay constant. It is not hard to see that $F_{\rho, \mu}(x, \sigma)$ is the continuous version of the entropy. Next, we extend the definition of σ_i by defining $\sigma_{\rho, \mu}^* : \mathbb{R}^d \rightarrow \mathbb{R}$ the unique solution of the equation

$$F_{\rho, \mu}(x, \sigma) = 0. \quad (1.16)$$

The introduction of σ^* aligns with the procedure of finding σ_i that controls the entropy to be the same for all i . Of course, the existence and uniqueness of σ^* is not trivial and will be discussed in section 5. For the symmetric probability mass, let $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$ be a given function and $p_\psi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be defined by

$$p_\psi(x, x') = \frac{1}{2} \left(\frac{\mathbf{Ker}_I(x, x', \psi(x))}{\int \mathbf{Ker}_I(x, x', \psi(x)) d\mu(x')} + \frac{\mathbf{Ker}_I(x, x', \psi(x'))}{\int \mathbf{Ker}_I(x, x', \psi(x')) d\mu(x)} \right). \quad (1.17)$$

For the extension on the output side, let $q : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}$ be given by

$$q(y, y') = \frac{\mathbf{Ker}_O(y, y')}{\iint \mathbf{Ker}_O(y, y') d\mu(y) d\mu(y')}. \quad (1.18)$$

Based on the construction of p and q , define

$$I_\rho(\mu) = \iint p_{\sigma_{\rho, \mu}^*}(x, x') \log \left(\frac{p_{\sigma_{\rho, \mu}^*}(x, x')}{q(y, y')} \right) d\mu(x, y) d\mu(x', y'). \quad (1.19)$$

Recall $\mu_X \in \mathcal{M}(\mathbb{R}^d)$ is an arbitrary given probability measure. We set $\mathcal{P}_X$ to be the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^s$ whose marginal coincides with μ_X . Formally speaking,

$$\mathcal{P}_X := \{\mu \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s) : \text{the } \mathbb{R}^d \text{ marginal of } \mu \text{ is } \mu_X\}. \quad (1.20)$$

Lastly, define

$$\tilde{\mathcal{P}}_X := \left\{ \mu \in \mathcal{P}_X : \int (p_{\sigma_{\rho, \mu}^*}(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} \mathbf{Ker}_O(y, y')}{\mathbf{Ker}_O(y, y')} d\mu(x', y') = 0, \mu(x, y)\text{-a.e.} \right\}. \quad (1.21)$$

2 Main Results

Using the established setup, we are able to prove the same results as in [Auffinger and Fletcher \[2023\]](#) with a broad class of input and output kernels. On top of that, compared to the original paper, we no longer require the sample probability measure μ_X to have C^1 density. We managed to relax the condition to C^0 .

Theorem 1. Assume μ_X has a continuous density $f(x)$ and compact support. Let $X_i \in \mathbb{R}^d$ be i.i.d random variables with probability measure μ_X . If condition 1 and condition 2 hold, then for any $\rho \in (0, 1)$,

$$\lim_{n \rightarrow \infty} \inf_Y L_{n,\rho}(X, Y) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu). \quad (2.1)$$

Moreover, there exist a measure μ^* and a sub-sequence $\{n_k\}$ such that $\mu_{n_k} \rightarrow \mu^*$ weakly and $I_\rho(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu)$.

Theorem 2. Under the same assumptions of theorem 1, μ^* has compact support.

Remark 2. In [Auffinger and Fletcher \[2023\]](#), several lemmas are proved under the assumption that the sample distribution μ_X is sub-Gaussian. This can also be done for our work, if we replace condition 1 with the following condition 3:

Condition 3 (Alternative Input Kernel Conditions). Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of $\mathbf{Ker}_I(x, x', \sigma) = \exp(-w(\sigma \|x - x'\|^\theta))$. Here, σ is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. Define

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \mid \exists \epsilon > 0, \text{ s.t. } \lim_{n \rightarrow \infty} n^{-1/2} \exp \left(6w \left(\left(\tilde{C} \sqrt{\log n} \right)^\theta \right) \right) = O(n^{-\epsilon}) \right\}. \quad (2.2)$$

Then w satisfies:

1. $\forall t \geq 0, w'(t) > 0. \lim_{t \rightarrow \infty} w(t) = \infty;$
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0;$
3. The following integral converges:

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty; \quad (2.3)$$

4. $\sup \mathcal{K}_w > 0$, and $\forall t \geq 0, w'(t) \leq w(t).$

We will not include proofs for lemmas assuming condition 3 and sub-Gaussian samples, because such generalization will not carry onto the main theorems. One could adapt the proofs we present assuming μ_X being sub-Gaussian and condition 3. However, it remains unknown that whether both theorem 1 and 2 hold assuming a more general data distribution.

3 The Generality of Conditions

Since the main goal of our work is to generalize the theoretical guarantees of the t-SNE algorithm, it is important that we do not lose comprehensiveness when assuming a concrete form of both the input and output kernels. Therefore, both condition 1 and 2 warrant an explanation.

For the formula of the input kernel in definition 1, we start by realizing that t -SNE is about the relationship between all data points. The insights we gain from the algorithm should be translation and rotation invariant with respect to the coordinate system we choose. Hence, it is only reasonable to assume that the input kernel is a function of $\|x - x'\|$ for each pair of input data points x, x' .

On the other hand, the weight we assign to each pair of data points is always positive due to the nature of probability masses. We thus lose no generality by assuming the input kernel

has an exponential form because obviously, the exponential function is a smooth bijection from $\mathbb{R}$ to $\mathbb{R}^+$. The same above can be said for the output kernel.

Additionally, though it is coincidental that in the original version of the algorithm σ_i can be viewed as the variance of a Gaussian random variable, in reality σ_i 's are only used to control the entropy of each probability distribution induced by x_i . With that being said, we only need to introduce σ in such a way that it is able to control the entropy for each x . As we can see in section 5, simply multiplying σ by $\|x - x'\|^\theta$ is enough to achieve this goal. Moreover, we have the convergence of σ^* in section 6.1, providing the anchor of the following results just as in Auffinger and Fletcher [2023].

Combining all the above, we arrive at the form of the input kernel in (1.4), by adding a general function w to the product $\sigma \|x - x'\|^\theta$. This gives us a kernel we can easily work on, without losing any generality.

Besides the form of the input kernel, we also want to make sure that the constraints we impose on the kernels in condition 1 and 2 are also general enough for the significance of our results. Since we frequently deal with integrals containing the input and output kernels, their integrability is the bare minimum we should assume. Along with integrability comes the requirement of their decay to 0 at infinity. Apart from that, we do not add any excessive constraints to the kernels and readers will see later in proofs that those assumptions are strictly necessary for the results to hold.

Specifically, notice the second assumption of condition 1 is basically requiring the input kernel to decay at least at a minimal speed by controlling its second order derivative. If we let $w(t) = \log t$, we have $w'(t) + tw''(t) = 0$. Roughly speaking, this implies that the main theorems hold as long as the input kernel decays at polynomial rate or faster, which is a big step up than proving the convergence for Gaussian kernels alone.

4 Summary and Structure of Proofs

Before we move onto mathematical details, we dedicate this short section to give a summary of our work and the structure of the proofs. Since t-SNE assigns a “weight” to each pair of data points using Gaussian and t -distribution, we propose the concept of generalized kernels so that the generalized t-SNE can use a wide range of functions to define the weights, thus the subsequent probability distributions associated with the input and output data points. A natural question to ask is: under which conditions can we prove the convergence of the generalized t-SNE algorithm towards an equilibrium measure? Our work here answers the questions by providing a concrete formulation of the generalized kernels as well as rigorous proofs of the convergence towards the equilibrium measure.

The generality of the kernel formulation as well as the conditions we impose are discussed in the previous section. For the proofs, we first establish the existence and uniqueness of σ^* as the asymptotic limit of σ_i 's in section 5. Then we state the uniform convergence of σ^* in section 6.1. σ^* serves as the anchor point of all subsequent convergence results. With the existence and uniqueness of σ^* as well as its convergence established, we are able to prove the convergence towards the equilibrium measure by proving inequalities from both sides, the details of which are discussed in section 6.2 and 6.3. Finally the proofs are wrapped up in section 6.5 and 6.6, where we show that the equilibrium measure also has compact support just as the input measure μ_X .

5 Existence and Uniqueness of σ^*

In this section we establish the existence and uniqueness of $\sigma_{\rho,\mu}^*(x)$. Recall that $\sigma_{\rho,\mu}^*(x)$ is the unique solution of the equation

$$F_{\rho,\mu}(x, \sigma) = \int \frac{\mathbf{Ker}_{\mathbf{I}}(x, x', \sigma)}{\int \mathbf{Ker}_{\mathbf{I}}(x, x', \sigma) d\mu(x')} \log \left(\frac{\mathbf{Ker}_{\mathbf{I}}(x, x', \sigma)}{\int \mathbf{Ker}_{\mathbf{I}}(x, x', \sigma) d\mu(x')} \right) d\mu(x') + \log \rho = 0 \quad (5.1)$$

Intuitively speaking, σ^* is the limit function of σ_i 's that are used to control the entropy. The purpose of this section is to prove that such limit exists and is unique for each given x . The existence and uniqueness of σ^* will become the foundation of everything we will prove afterwards. By condition 1, we can write

$$\begin{aligned} F_{\rho,\mu}(x, \sigma) &= \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \log \left(\frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \right) d\mu(x') + \log \rho \\ &= - \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\ &\quad - \log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) + \log \rho. \end{aligned} \quad (5.2)$$

We start with the lemma that gives F an increasing lower bound.

Lemma 1. *If μ has a continuous, bounded density $f(x)$, then there exists a constant $C > 0$ such that*

$$F_{\rho,\mu}(x, \sigma) \geq C \log \sigma - \log f(x) + \log \rho. \quad (5.3)$$

Proof. Notice that for any integrable function h on $\mathbb{R}_{\geq 0}$, we have

$$\int_{\mathbb{R}^d} h(\|x\|) dx = \int_{\mathbb{S}^{d-1}} \int_0^\infty r^{d-1} h(r) dr d\nu = S_{d-1} \int_0^\infty r^{d-1} h(r) dr, \quad (5.4)$$

where $\mathbb{S}^{d-1}$ is the $(d-1)$ -dimensional sphere, $d\nu$ the corresponding surface measure, and S_{d-1} the volume of the sphere. By Condition 1, we know that

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (5.5)$$

Since w is increasing and $\lim_{t \rightarrow \infty} w(t) = \infty$, it is easy to see that

$$\int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt < \infty. \quad (5.6)$$

Denote

$$Z_d := \int_{\mathbb{R}^d} \exp(-w(\|\tau\|^\theta)) d\tau = S_{d-1} \int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt. \quad (5.7)$$

Consider

$$\begin{aligned} \int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') &= \int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx' \\ &= Z_d \sigma^{-d/\theta} \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{Z_d \sigma^{-d/\theta}} f(x') dx'. \end{aligned} \quad (5.8)$$

Because $\lim_{x \rightarrow \infty} w(x) = \infty$, the integrand of the above integral converges weakly to $\delta_x f(x')$. To prove this statement, notice that

$$\begin{aligned}
& \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{Z_d \sigma^{-d/\theta}} f(x') dx' \\
&= \frac{1}{Z_d} \int \exp\left(-w\left(\|\sigma^{1/\theta}(x - x')\|^\theta\right)\right) f(x') d\sigma^{1/\theta}(x - x') \\
&\equiv \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau.
\end{aligned} \tag{5.9}$$

Since f is bounded, $\exp(-w(\|\tau\|^\theta)) \sup f$ is integrable. By Dominant Convergence Theorem and the continuity of f , we have

$$\begin{aligned}
& \lim_{\sigma \rightarrow \infty} \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau \\
&= \frac{1}{Z_d} \int \lim_{\sigma \rightarrow \infty} \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau \\
&= f(x) \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) d\tau \\
&= f(x).
\end{aligned} \tag{5.10}$$

There exists some $\epsilon(\sigma) \rightarrow 0$ as $\sigma \rightarrow \infty$ and a constant $C > 0$, that

$$\begin{aligned}
-\log\left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')\right) &= -\log\left(Z_d \sigma^{-d/\theta}\right) - \log f(x) + \epsilon(\sigma) \\
&\geq C \log \sigma - \log f(x).
\end{aligned} \tag{5.11}$$

Next, since f is bounded, there exists $M > 0$ such that $f \leq M$.

$$\begin{aligned}
& \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\
&\leq M \frac{\int w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta)) dx'}{\int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx'} \\
&= M \frac{\sigma^{-d/\theta} \int w(\|\sigma^{1/\theta}(x - x')\|^\theta) \exp(-w(\|\sigma^{1/\theta}(x - x')\|^\theta)) d\sigma^{1/\theta}(x - x')}{\int \exp(-w(\sigma \|(x - x')\|^\theta)) f(x') dx'} \\
&= M \frac{\sigma^{-d/\theta} \int w(\|\tau\|^\theta) \exp(-w(\|\tau\|^\theta)) d\tau}{\int \exp(-w(\sigma \|(x - x')\|^\theta)) f(x') dx'} \\
&\rightarrow \frac{M}{Z_d f(x)} \int w(\|\tau\|^\theta) \exp(-w(\|\tau\|^\theta)) d\tau < \infty.
\end{aligned} \tag{5.12}$$

The numerator converges because of condition 1. As the bound in (5.12) does not depend on σ , we can combine (5.2), (5.11) and (5.12) and deduce that

$$F_{\rho, \mu}(x, \sigma) \geq C \log \sigma - \log f(x) + \log \rho. \tag{5.13}$$

□

This lemma proves the existence of $\sigma_{\rho, \mu}^*(x)$. Indeed, by (5.2), apparently

$$\lim_{\sigma \rightarrow 0} F_{\rho, \mu}(x, \sigma) = \log \rho < 0 \tag{5.14}$$

since $\rho \in (0, 1)$. Now with lemma 1, we get

$$\lim_{\sigma \rightarrow \infty} F_{\rho, \mu}(x, \sigma) = \infty. \quad (5.15)$$

By continuity of f and intermediate value theorem, $\sigma_{\rho, \mu}^*(x)$ exists. To prove the uniqueness, we are left to show $F_{\rho, \mu}$ is strictly increasing. The next lemma will do just that.

Lemma 2. *Given the same condition as in lemma 1, $F_{\rho, \mu}(x, \sigma)$ is strictly increasing in σ .*

To tackle this proof, we need one additional lemma inspired by theorem 7 in Liu et al. [2009].

Lemma 3. *Let $f(x), g(x), h(x)$ be continuous functions $\mathbb{R}^d \rightarrow \mathbb{R}^+$ such that for all $x, y \in \mathbb{R}^d$*

$$(g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) \geq 0. \quad (5.16)$$

Let μ be an arbitrary Lebesgue measure and assume all integrals in this lemma converge. Then the inequality

$$\frac{\int_{\mathbb{R}^d} f(x) d\mu}{\int_{\mathbb{R}^d} h(x) d\mu} \geq \frac{\int_{\mathbb{R}^d} f(x) g(x) d\mu}{\int_{\mathbb{R}^d} h(x) g(x) d\mu} \quad (5.17)$$

holds. If (5.16) reverses, then (5.17) reverses.

Proof. The proof of this lemma is similar to the one in Liu et al. [2009]. It suffices to show that

$$\int_{\mathbb{R}^d} f(x) d\mu \int_{\mathbb{R}^d} h(x) g(x) d\mu \geq \int_{\mathbb{R}^d} h(x) d\mu \int_{\mathbb{R}^d} f(x) g(x) d\mu. \quad (5.18)$$

The above inequality is equivalent to

$$\int_{\mathbb{R}^d} f(x) d\mu(x) \int_{\mathbb{R}^d} h(y) g(y) d\mu(y) \geq \int_{\mathbb{R}^d} h(x) d\mu(x) \int_{\mathbb{R}^d} f(y) g(y) d\mu(y). \quad (5.19)$$

That means we need to show

$$\mathcal{G} := \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g(y) (f(x) h(y) - f(y) h(x)) d\mu(x) d\mu(y) \geq 0. \quad (5.20)$$

Swap x and y in the formula of $\mathcal{G}$ and we have an equivalent expression:

$$\mathcal{G} = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g(x) (f(y) h(x) - f(x) h(y)) d\mu(x) d\mu(y). \quad (5.21)$$

Adding the last two formulas together yields

$$\begin{aligned} 2\mathcal{G} &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (g(x) - g(y)) (f(y) h(x) - f(x) h(y)) d\mu(x) d\mu(y) \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} h(x) h(y) (g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) d\mu(x) d\mu(y) \geq 0 \end{aligned} \quad (5.22)$$

by the hypothesis of the lemma. This finishes the proof. $\square$

Proof of lemma 2. We compute the derivative of $F_{\rho, \mu}(x, \sigma)$ directly using (5.2). The following simplified notations will be used in the first several steps:

$$w = w(\sigma \|x - x'\|^\theta), \quad w' = w'(\sigma \|x - x'\|^\theta), \quad \exp(\cdot) = \exp(-w(\sigma \|x - x'\|^\theta)). \quad (5.23)$$

Omitting $d\mu(x')$ in the integration for cleaner writing, and we have

$$\begin{aligned}
& \frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) \\
&= - \int \frac{w' \|x - x'\|^\theta \exp(\cdot)}{\int \exp(\cdot)} - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} - \frac{(\int \exp(\cdot))'}{\int \exp(\cdot)} \\
&= - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} \quad (\text{The first and fourth terms cancel.}) \\
&= \int \frac{w \cdot w' \cdot \|x - x'\|^\theta \cdot \exp(\cdot)}{\int \exp(\cdot)} - \frac{\int w \cdot \exp(\cdot) \int w' \cdot \|x - x'\|^\theta \exp(\cdot)}{(\int \exp(\cdot))^2}. \tag{5.24}
\end{aligned}$$

We are able to move differentiation inside the integral because of dominant convergence theorem. To show that $\frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) \geq 0$, it suffices to show

$$\int w \cdot w' \cdot \|x - x'\|^\theta \cdot \exp(\cdot) \int \exp(\cdot) \geq \int w \cdot \exp(\cdot) \int w' \cdot \|x - x'\|^\theta \exp(\cdot). \tag{5.25}$$

For each $x, y, x' \in \mathbb{R}$ if the equality

$$\begin{aligned}
& (w'(\sigma \|x - x'\|^\theta) \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \|y - x'\|^\theta) \\
& \cdot \left(\frac{1}{w(\sigma \|y - x'\|^\theta)} - \frac{1}{w(\sigma \|x - x'\|^\theta)} \right) \geq 0 \tag{5.26}
\end{aligned}$$

is true, then setting $f(x) = \exp(\cdot)$, $h(x) = w \cdot \exp(\cdot)$ and $g(x) = w' \cdot \|x - x'\|^\theta$ in lemma 3 shows that (5.25) holds. Let $\|x - x'\|^\theta > \|y - x'\|^\theta$ without loss of generality. Then by monotonicity of w , we have $w(\sigma \|x - x'\|^\theta) > w(\sigma \|y - x'\|^\theta)$. For (5.26) to be true, we are left to show

$$w'(\sigma \|x - x'\|^\theta) \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \|y - x'\|^\theta \geq 0. \tag{5.27}$$

Since $\sigma > 0$, the above inequality is equivalent to:

$$w'(\sigma \|x - x'\|^\theta) \sigma \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \sigma \|y - x'\|^\theta \geq 0. \tag{5.28}$$

Let $\tilde{w}(x) = xw'(x)$. Based on the second assumption in Condition 1 and mean value theorem, $\tilde{w}$ satisfies for all $a, b \geq 0$, $a > b$, there exists $c \in [b, a]$ that

$$\frac{aw'(a) - bw'(b)}{a - b} = w'(c) + cw''(c) \geq 0. \tag{5.29}$$

This proves (5.28) by letting $a = \sigma \|x - x'\|^\theta$ and $b = \sigma \|y - x'\|^\theta$, as desired. $\square$

We can now formally state and prove the main result of this section.

Proposition 1. *Let $\mu \in \mathcal{M}(\mathbb{R}^d)$ with C^0 density f such that f is bounded. The equation $F_{\rho, \mu}(x, \sigma) = 0$ has a unique solution $\sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. Moreover, $x \mapsto \sigma_{\rho, \mu}^*(x)$ is smooth.*

Proof. The existence and uniqueness can be proven by combining (5.14), (5.15), lemma 1 and lemma 2. The smoothness of $\sigma_{\rho, \mu}^*(x)$ follows from the implicit function theorem. $\square$

6 Proofs of the Main Theorems

From now on, we will assume both condition 1 and condition 2 throughout this section. We will also assume that the distribution μ_X that we draw samples from has C^0 density with bounded density and gradient of density. The assumptions on the sample distribution μ_X are inherited directly from section 5 to avoid lengthy theorem and lemma statements.

6.1 Convergence Lemmas for Input Kernels

The main difficulties of proving the convergence lemmas related to the input kernel is the existence and uniqueness of $\sigma_{\rho,\mu}^*(x)$, which we have already established in section 5. The rest of lemmas are similar, if not easier than the corresponding results in [Auffinger and Fletcher \[2023\]](#). We will list the important lemmas here without proofs. The reader can easily adapt the proofs in [Auffinger and Fletcher \[2023\]](#) to our case.

First we have the two following important propositions about the uniform convergence of $F_{\mu_n}(x, \sigma)$ and $\sigma_{\mu_n}^*(x)$.

Proposition 2. *Let μ_X be a probability measure with compact support, and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . $\mu_n = (1/n) \sum_i \delta_{X_i}$ is the empirical measure. Then there exists a constant $\epsilon > 0$, that almost surely in $X_1, \dots, X_n$,*

$$\sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} |F_{\mu_n}(X_i, \sigma) - F_\mu(X_i, \sigma)| = O(n^{-\epsilon}). \quad (6.1)$$

Proposition 3. *Suppose μ_X has compact support and C^0 density. Then almost surely, as $n \rightarrow \infty$,*

$$\sup_{i=1, \dots, n} |\sigma_{\mu_n}^*(X_i) - \sigma_\mu^*(X_i)| \rightarrow 0. \quad (6.2)$$

In particular,

$$|\sigma_{\mu_n}^*(X_i) - \sigma_\mu^*(X_i)| = O(|F_{\mu_n}(X_i, \sigma) - F_\mu(X_i, \sigma)|). \quad (6.3)$$

Next two lemmas can be borrowed directly from [Auffinger and Fletcher \[2023\]](#) as they are irrelevant to the generalized kernels.

Lemma 4. *There exists μ^* such that*

$$I_\rho(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu) \quad (6.4)$$

To further simplify up the notations, define

$$\begin{aligned} \sigma_n(x) &= \sigma_{\rho, \mu_n}^*(x), \sigma(x) = \sigma_{\rho, \mu_X}^*(x), \\ f(x, x') &= f_\sigma(x, x') =: \exp(-w(\sigma \|x - x'\|^\theta)), f_n(x, x') = f_{\sigma_n}(x, x'), \\ p'_n(x, x') &= \frac{f_n(x, x')}{\int f_n(x, x') d\mu_n(x') - 1/n}, \quad p'(x, x') = \frac{f(x, x')}{\int f(x, x') d\mu(x')}, \\ p_n(x, x') &= \frac{1}{2}(p'_n(x, x') + p'_n(x', x)), \quad p(x, x') = \frac{1}{2}(p'(x, x') + p'(x', x)). \end{aligned} \quad (6.5)$$

Note that $p_{j|i} = p'_n(X_i, X_j)/n$ and $p_{ij} = p_n(X_i, X_j)/n^2$, and for $\sigma^{-1} \in [\delta^\theta, n^{\theta/2}]$ we have

$$\int f(x, x') d\mu(x'), \quad \int f_n(x, x') d\mu_n(x') \geq M \quad (6.6)$$

for some constant $M > 0$.

The next few lemmas cover the rest of convergence results for the input kernels. The proofs can be easily adapted from [Auffinger and Fletcher \[2023\]](#). We omit them here for brevity.

Lemma 5. *Let μ_X have compact support. There exists $\epsilon > 0$ such that as $n \rightarrow \infty$,*

$$\begin{aligned} \sup_{i,j} |p'_n(X_i, X_j) - p'(X_i, X_j)| &= O(n^{-\epsilon}), \\ \sup_{i,j} |p_n(X_i, X_j) - p(X_i, X_j)| &= O(n^{-\epsilon}). \end{aligned}$$

Lemma 6. *As $n \rightarrow \infty$, we have*

$$\sum_{i \neq j} p_{ij} \log p_{ij} = \iint p(x, x') \log p(x, x') d\mu d\mu - 2 \log n + o(1). \quad (6.7)$$

6.2 Lower Bound for $\inf_{Y \in (\mathbb{R}^s)^n} L_{n,\rho}(X, Y)$

We write $\mathbf{Ker}_O(y, y') = g(y, y') = k(\|y - y'\|)$ from now on for simpler notations.

Lemma 7. *There exists a set of points $y^*(X)$ such that*

$$\inf_{Y \in (\mathbb{R}^s)^n} L_{n,\rho}(X, Y) = L_{n,\rho}(X, y^*). \quad (6.8)$$

Proof. Let $k \in \{1, \dots, n\}$. We are going to show that

$$\lim_{|y_k| \rightarrow \infty} L_{n,\rho}(X, y) = \infty. \quad (6.9)$$

By definition, L is the relative entropy and

$$L_{n,\rho}(X, y) = \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \quad (6.10)$$

For n given and fixed, it suffices to show that

$$\sum_{i \neq j} p_{ij} \log(q_{ij}) = \sum_{i \neq j} p_{ij} \log \left(\frac{g(y_i, y_j)}{\sum_{m \neq l} g(y_m, y_l)} \right) \rightarrow -\infty. \quad (6.11)$$

Recall that $g(y, y')$ is bounded by condition 2. We have

$$\begin{aligned} & - \sum_{i \neq j} p_{ij} \log \left(\sum_{m \neq l} g(y_m, y_l) \right) = - \log \left(\sum_{m \neq l} g(y_m, y_l) \right) \\ & \leq - \sum_{i \neq j} p_{ij} \log \left(\sum_{m \neq k, l \neq k, m \neq l} g(y_m, y_l) \right) < \infty. \end{aligned} \quad (6.12)$$

On the other hand, we have $g(y_i, y_k) = k(\|y_i - y_k\|) \rightarrow 0$ as $|y_k| \rightarrow \infty$ because $\|y_i - y_k\| \rightarrow \infty$ and k satisfies

$$\iint_{(0, \infty)^2} k(\|x - x'\|) dx dx' < \infty. \quad (6.13)$$

This means

$$\lim_{|y_k| \rightarrow \infty} \sum_{i \neq j} p_{ij} \log(g(y_i, y_j)) \rightarrow -\infty. \quad (6.14)$$

Hence, the minimum of the $L_{n,\rho}(X, y)$ is achieved on a bounded set, the lemma follows by the continuity of $L_{n,\rho}(X, y)$. $\square$

Before we move on, we would like to take a small deviation and compute the partial derivative of the relative entropy. This is important because it will impact the property of the set $\tilde{\mathcal{P}}_X$.

Lemma 8. *Let $X \in (\mathbb{R}^d)^n$ and $Y = (y_1, \dots, y_n) \in (\mathbb{R}^s)^n$ be the input and output of the t -SNE algorithm, then we have*

$$\frac{\partial L_{n,\rho}(X, Y)}{\partial y_i} = - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}. \quad (6.15)$$

Proof. By definition,

$$\begin{aligned} L_{n,\rho}(X, Y) &= \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) \\ &= \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \end{aligned} \quad (6.16)$$

The first summation does not depend on y_i and will vanish when taking the partial derivative. Moreover, we can further breakdown the second summation:

$$\begin{aligned} - \sum_{i \neq j} p_{ij} \log q_{ij} &= - \sum_{i \neq j} p_{ij} \log g(y_i, y_j) + \sum_{i \neq j} p_{ij} \log \left(\sum_{k \neq l} g(y_k, y_l) \right) \\ &= - \sum_{i \neq j} p_{ij} \log g(y_i, y_j) + \log \left(\sum_{k \neq l} g(y_k, y_l) \right). \end{aligned} \quad (6.17)$$

Taking derivative of the last expression and note that only terms that contain y_i remain. This means

$$\begin{aligned} \frac{\partial L_{n,\rho}(X, Y)}{\partial y_i} &= \sum_{j \neq i, j=1}^n \left(-p_{ij} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} + \frac{1}{\sum_{k \neq l} g(y_k, y_l)} \frac{\partial}{\partial y_i} g(y_i, y_j) \right) \\ &= \sum_{j \neq i, j=1}^n \left(-p_{ij} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} + \frac{g(y_i, y_j)}{\sum_{k \neq l} g(y_k, y_l)} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} \right) \\ &= - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}. \end{aligned} \quad (6.18)$$

$\square$

We can easily do a sanity test by letting $g(y_i, y_j) = (1 + \|y_i - y_j\|^2)^{-1}$ and get formula (A.1) in [Auffinger and Fletcher \[2023\]](#). The derivative in lemma 8 explains why $\tilde{\mathcal{P}}_X$ has that specific form.

Lemma 9. *Assume μ_X has compact support. The following holds:*

$$\liminf_{n \rightarrow \infty} \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) \geq \iint p(x, x') \log \frac{p(x, x')}{q(y, y')} d\mu d\mu. \quad (6.19)$$

Proof. Just as in lemma 8, we first break down the summation:

$$\sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \quad (6.20)$$

The convergence of the first summation has been established in lemma 6. We just need to deal with the second summation. Note that we now have

$$\mu_n = \frac{1}{n} \sum \delta_{X_i} \times \delta_{y_i} \quad (6.21)$$

if y_i is involved. By the definition of q_{ij} , we have

$$\begin{aligned} & - \sum_{i \neq j} p_{ij} \log(q_{ij}) \\ &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log g(y_i, y_j) + \log \left(\frac{1}{n^2} \sum_{i \neq j} g(y_i, y_j) \right) + 2 \log n \\ &= \iint p_n(x, x') g(y, y') d\mu_n d\mu_n + \log \left(\iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \right) + 2 \log n. \end{aligned} \quad (6.22)$$

By Fatou's lemma and lemma 5, we have

$$\liminf_{n \rightarrow \infty} \iint p_n(x, x') \log g(y, y') d\mu_n d\mu_n \geq \iint p(x, x') \log g(y, y') d\mu d\mu. \quad (6.23)$$

Similarly, using Fatou lemma again and we have

$$\liminf_{n \rightarrow \infty} \iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \geq \iint g(y, y') d\mu d\mu. \quad (6.24)$$

The $1/n$ is the duplicated diagonal term in the empirical measure and will obviously vanish as we take $n \rightarrow \infty$. Moreover, because of condition 2, the integrals on the RHS of the last two inequalities converge. In addition, since $\log$ is a continuous and increasing function, (6.24) implies

$$\liminf_{n \rightarrow \infty} \log \left(\iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \right) \geq \log \left(\iint g(y, y') d\mu d\mu \right). \quad (6.25)$$

The proof is finished by combining (6.22), (6.23), (6.25) and lemma 6. The $2 \log n$ cancelled out because of lemma 6. □

Proposition 4. *There exists a sub-sequence $n(k)$ such that*

$$\liminf_{k \rightarrow \infty} \inf_{Y \in (\mathbb{R}^s)^{n(k)}} L_{n(k), \rho}(X, Y) \geq I_\rho(\mu). \quad (6.26)$$

Proof. Let y^* be given in the lemma and define

$$\mu_n = \frac{1}{n} \sum \delta_{X_i} \times \delta_{y_i}. \quad (6.27)$$

We can easily show that μ_n converges weakly to a measure μ . By lemma 14 in section 6.4, we know that the limiting measure $\mu \in \mathcal{P}_X$. The proof is complete by using lemma 7 and lemma 9. □

6.3 Upper Bound for $\inf_{Y \in (\mathbb{R}^s)^n} L_{n,\rho}(X, Y)$

For convenience, define

$$q_n(y, y') = \frac{g(y, y')}{\iint g(y, y') d\mu_n(y) d\mu_n(y') - \frac{1}{n} k(0)}. \quad (6.28)$$

Here $k(0) = g(y, y)$ is the value of the output kernel when both inputs are equal. Since multiplying a constant to g does not change anything, we can just treat $k(0) = 1$ without loss of generality as everything is normalized. By lemma 4 there exists μ^* such that $I(\mu^*) = \inf_{\mu} I_{\rho}(\mu)$. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. random variables drawn from μ^* and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{(X_i, Y_i)}$ their empirical measure. μ_n converges to μ^* weakly.

Lemma 10. *As $n \rightarrow \infty$, we have*

$$\sum_{i \neq j} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \iint p_n(x, x') \log \left(\frac{p_n(x, x')}{q_n(y, y')} \right) d\mu_n d\mu_n + o(1). \quad (6.29)$$

Proof. We begin with the familiar computation

$$\begin{aligned} \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log \left(\frac{p_n(X_i, X_j)}{q_n(X_i, X_j)} \right) \\ &= \iint p_n(x, x') \log \left(\frac{p_n(x, x')}{q_n(x, x')} \right) d\mu_n d\mu_n + \text{Err}_n, \end{aligned} \quad (6.30)$$

where

$$\begin{aligned} \text{Err}_n &= -\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \left(\log \left(\frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \right) - \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right) \right) \\ &= -\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \right) \\ &\quad + \frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right). \end{aligned} \quad (6.31)$$

By bounds in (6.6), it is easy to see that the first summation in the error term is $o(1)$ and we can write

$$\text{Err}_n = o(1) + \frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right). \quad (6.32)$$

For the second term, it is already clear that

$$\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} = o(1). \quad (6.33)$$

On the other hand, we have by Fatou's lemma:

$$\liminf_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k \neq l} g(Y_k, Y_l) = \liminf_{n \rightarrow \infty} \iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \geq \iint g(y, y') d\mu^* d\mu^* < \infty, \quad (6.34)$$

by condition 2. We can see that the second term in Err_n is also $o(1)$, which immediately implies $\text{Err}_n = o(1)$ and completes the proof. $\square$

The next lemma is needed to close up the lower bound part of the main theorem.

Lemma 11. *As $n \rightarrow \infty$, we have*

$$\iint g(y, y') d\mu_n(y) d\mu_n(y') - \frac{1}{n} = \iint g(y, y') d\mu^* d\mu^* + o(1). \quad (6.35)$$

Proof. Since $\mu_n \rightarrow \mu$ weakly, it is tight by Prokhorov's theorem. For arbitrary $\epsilon > 0$, there exists a compact set K such that for all n , $\mu^*(K^c), \mu_n(K^c) \leq \epsilon$. Since $g(y, y')$ is bounded, there exists a constant C such that

$$\begin{aligned} & \left| \iint g(y, y') d\mu_n(y) d\mu_n(y') - \iint g(y, y') d\mu^* d\mu^* \right| \\ &= \left| \iint g(y, y') \mathbb{1}_{K \times K} (d\mu_n d\mu_n - d\mu^* d\mu^*) \right| + C\epsilon. \end{aligned} \quad (6.36)$$

Define

$$G_n(y) = \int g(y, y') \mathbb{1}_K d\mu_n(y'), \quad G(y) = \int g(y, y') \mathbb{1}_K d\mu^*(y'). \quad (6.37)$$

We now require the kernel g to have bounded derivative, and we have the following claims:

Claim 1. *$G_n(y)$ is uniformly equicontinuous on K .*

Proof of Claim. By mean value theorem and condition 2, we get

$$\begin{aligned} |G_n(y) - G_n(\tilde{y})| &= \left| \int k(\|y - y'\|) - k(\|\tilde{y} - y'\|) \mathbb{1}_K d\mu_n(y') \right| \\ &\leq \int \|y - \tilde{y}\| \left| \frac{k(\|y - y'\|) - k(\|\tilde{y} - y'\|)}{\|(y - y') - (\tilde{y} - y')\|} \right| \mathbb{1}_K d\mu_n(y') \\ &= O(\|y - \tilde{y}\|). \end{aligned} \quad (6.38)$$

□

Claim 2. $\sup_{y \in K} |G_n(y) - G(y)| = o(1)$.

Proof of Claim. Since K is compact, it suffices to show that for each $y \in K$, there exists $\delta > 0$ such that

$$\sup_{\tilde{y} \in B(y, \delta)} |G_n(\tilde{y}) - G(\tilde{y})| \leq \epsilon. \quad (6.39)$$

Towards this end, by triangular inequality,

$$|G_n(\tilde{y}) - G(\tilde{y})| \leq |G_n(\tilde{y}) - G_n(y)| + |G_n(y) - G(y)| + |G(y) - G(\tilde{y})|. \quad (6.40)$$

By the previous claim, the first term on the RHS vanishes because of equicontinuity. The second term vanishes because of weak convergence and the boundedness of the integrand $g(y, y')$. Finally, the last term vanishes because we can prove the equicontinuity of G using a similar argument like the claim 1.

□

By claim 2 we have

$$\iint g(y, y') \mathbb{1}_{K \times K} d\mu_n d\mu_n = \iint g(y, y') \mathbb{1}_{K \times K} d\mu^* d\mu^* + o(1) = \int G(y) \mathbb{1}_K d\mu_n(y) + o(1). \quad (6.41)$$

Since G is bounded and continuous by condition 2, using weak convergence of μ_n and we have

$$\int G(y) \mathbb{1}_K d\mu_n(y) = \int G(y) \mathbb{1}_K d\mu^*(y) + o(1) = \iint g(y, y') d\mu^* d\mu^* + o(1), \quad (6.42)$$

as desired. $\square$

Lemma 12. *If μ^* has compact support in Y , then*

$$\iint p_n(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p_n(x, x') \log(g(y, y')) d\mu^* d\mu^* + o(1). \quad (6.43)$$

Proof. By compactness, $\|y - y'\|$ is bounded from above and $g(y, y')$ does not decay to zero. In that case, by lemma 5, we can replace the p_n in the integral by p :

$$\iint p_n(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p(x, x') \log(g(y, y')) d\mu_n d\mu_n + o(1). \quad (6.44)$$

mimicking the proof of lemma 6 we can show that

$$\iint p(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p(x, x') \log(g(y, y')) d\mu^* d\mu^* + o(1). \quad (6.45)$$

This finishes the proof. $\square$

Lemma 13. *If μ^* has compact support in Y , then*

$$\sum_{i \neq j} p_{ij} \log\left(\frac{p_{ij}}{q_{ij}}\right) = \iint p(x, x') \log\left(\frac{p(x, x')}{q(y, y')}\right) d\mu^* d\mu^* + o(1) \quad (6.46)$$

Proof. This statement can be proved by combining the last 3 lemmas and lemma 6. $\square$

Using the previous lemma and we can immediately get

Proposition 5. *If μ^* has compact support in Y , then*

$$\limsup_{n \rightarrow \infty} \inf_{Y \in (\mathbb{R}^s)^n} L_{n, \rho}(X, Y) \leq I_\rho(\mu^*). \quad (6.47)$$

6.4 Auxiliary Lemmas

This section contains lemmas that are necessary but are not a part of the flow in the proofs of convergence. Recall that $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i, Y_i}$ is the empirical measure of both the input and output data points. We just ignore the first or the second part based on the integration variable because it is a probability measure. Let $Y = Y^* = (Y_1, \dots, Y_n)$ be a minimizer of L . Recall from (6.18) that

$$\frac{\partial L_{n, \rho}(X, Y)}{\partial y_i} = - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} = 0. \quad (6.48)$$

Lemma 14. *Suppose that μ_X has either compact support and a C^0 density $f(x)$. Then for μ -a.e. x and y we have*

$$\int (p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') = 0. \quad (6.49)$$

Proof. Let $\alpha(x, y)$ be a test function. It suffices to show that

$$\iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\partial_y g(y, y')}{g(y, y')} d\mu(x', y') = 0. \quad (6.50)$$

Recalling the definition of p_n and q_n , we see from (6.18) that for all i ,

$$\frac{1}{n^2} \sum_{j=1, j \neq i}^n (p_n(X_i, X_j) - q_n(X_i, X_j)) \frac{\partial_{Y_i} g(Y_i, Y_j)}{g(Y_i, Y_j)} = 0. \quad (6.51)$$

Multiply by $\alpha(X_i, Y_i)$ for each i and the both sides are still zero. Afterwards, take the summation over i and we get

$$\begin{aligned} & \iint \alpha(x, y)(p_n(x, x') - q_n(y, y')) \frac{\partial_y g(y, y')}{g(y, y')} d\mu_n d\mu_n \\ &= \frac{1}{n^2} \sum_{1 \leq i, j \leq n} (p_n(X_i, X_j) - q_n(X_i, X_j)) \frac{\partial_{Y_i} g(Y_i, Y_j)}{g(Y_i, Y_j)} = 0 \end{aligned} \quad (6.52)$$

if we require $k'(0) = 0$. By lemma 5 and lemma 11, we have

$$p_n(X_i, X_j) - q_n(Y_i, Y_j) = p(X_i, X_j) - q(Y_i, Y_j) + o(1). \quad (6.53)$$

Since both α and $\frac{\partial_{Y_i} g(Y_i, Y_j)}{g(Y_i, Y_j)}$ are bounded based on our assumptions, we have

$$\begin{aligned} & \iint \alpha(x, y)(p_n(x, x') - q_n(y, y')) \frac{\partial_y g(y, y')}{g(y, y')} d\mu_n d\mu_n \\ &= \iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\partial_y g(y, y')}{g(y, y')} d\mu_n d\mu_n + o(1). \end{aligned} \quad (6.54)$$

By lemma 15 and lemma 16 below we have

$$\begin{aligned} & \iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\partial_y g(y, y')}{g(y, y')} d\mu_n d\mu_n \\ &= \iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\partial_y g(y, y')}{g(y, y')} d\mu d\mu + o(1). \end{aligned} \quad (6.55)$$

Since α is an arbitrary test function,

$$\int (p(x, x') - q(y, y')) \frac{\partial_y g(y, y')}{g(y, y')} d\mu(x', y') = 0. \quad (6.56)$$

□

Lemma 15. *The following holds as $n \rightarrow \infty$:*

$$\iint \alpha(x, y)p(x, x') \frac{\partial_y g(y, y')}{g(y, y')} d\mu_n d\mu_n = \iint \alpha(x, y)p(x, x') \frac{\partial_y g(y, y')}{g(y, y')} d\mu d\mu + o(1) \quad (6.57)$$

Lemma 16. *The following holds as $n \rightarrow \infty$:*

$$\iint \alpha(x, y)q(y, y') \frac{\partial_y g(y, y')}{g(y, y')} d\mu_n d\mu_n = \iint \alpha(x, y)q(y, y') \frac{\partial_y g(y, y')}{g(y, y')} d\mu d\mu + o(1) \quad (6.58)$$

Both lemmas above can be proved via a similar strategy in the proof of lemma 10 by analyzing the diagonal terms.

6.5 Proof of Theorem 1

Write

$$d_{n,\rho}(X) = \inf_{Y \in (\mathbb{R}^s)^n} L_{n,\rho}(X, Y). \quad (6.59)$$

By proposition 5, 4 and the fact that $\mu^* = \arg \min_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu)$, there exists $\mu \in \tilde{\mathcal{P}}_X$ and a subsequence $n(k)$ such that

$$I_\rho(\mu^*) \geq \limsup_{n \rightarrow \infty} d_{n,\rho}(X) \geq \limsup_{n \rightarrow \infty} d_{n(k),\rho}(X) \geq I_\rho(\mu) \geq I_\rho(\mu^*). \quad (6.60)$$

Therefore the limit

$$\lim_{n \rightarrow \infty} d_{n(k),\rho}(X) = I_\rho(\mu) = I_\rho(\mu^*) \quad (6.61)$$

holds when we look at the sub-sequence $n(k)$ with the empirical measure μ converging to μ weakly. For the general limit, we can repeat the reasoning above by looking at any convergent sub-sequences of $d_{n,\rho}(X)$. We are able to find a sub-sequence inside each convergent sub-sequence and they all converges to $I_\rho(\mu^*)$. Therefore each convergent sub-sequence of $d_{n,\rho}(X)$ has the same limit and the theorem is proved. $\square$

6.6 Proof of Theorem 2

We need another lemma before proving this theorem.

Lemma 17. *For every $\mu \in \tilde{\mathcal{P}}_X$, there exists a function $h_\mu(x)$ that does not depend on y such that*

$$\int p(x, x') \log(1 + \|y - y'\|) d\mu = h_{\mu(x)} - \frac{\int g(y, y') d\mu}{\iint g(y, y') d\mu d\mu}. \quad (6.62)$$

Moreover, if μ_X has compact support, then h_μ is continuous on the support of μ_X .

Proof. Recall from lemma 8 and lemma 14, we have

$$\begin{aligned} 0 &= \int (p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') = \int p(x, x') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu - \frac{\int \frac{\partial}{\partial y} g(y, y') d\mu}{\iint g(y, y') d\mu d\mu} \\ &= \nabla_y \left(\int p(x, x') \log(g(y, y')^{-1}) d\mu - \frac{\int g(y, y') d\mu}{\iint g(y, y') d\mu d\mu} \right). \end{aligned} \quad (6.63)$$

This means that there exists $h_\mu : \mathbb{R}^d \rightarrow \mathbb{R}$ such that

$$h_{\mu(x)} = \int p(x, x') \log(g(y, y')^{-1}) d\mu + \frac{\int g(y, y') d\mu}{\iint g(y, y') d\mu d\mu}. \quad (6.64)$$

Since $h_{\mu(x)}$ does not change with y , it suffices to show that $\int p(x, x') \log(g(y, y')^{-1}) d\mu$ is continuous in x . Let $\epsilon > 0$ and suppose $x_n \rightarrow x$. Since p is continuous, and μ_X has compact support, we have for some constant C ,

$$\iint \log(g(y, y')^{-1}) d\mu d\mu \leq C \int p(x, x') \log(g(y, y')^{-1}) d\mu < \infty. \quad (6.65)$$

In particular, for almost every y , $\int \log(g(y, y')^{-1}) d\mu < \infty$. Since $p(x, x')$ is continuous and μ_X has compact support, $p(x, x')$ is uniformly continuous. Specifically, there exists $\delta = \delta(y, \mu)$ such that if $\|x_n - x\| \leq \delta$,

$$\sup_x |p_n(x, x') - p(x, x')| \leq \frac{\epsilon}{\int \log(g(y, y')^{-1}) d\mu}. \quad (6.66)$$

This means

$$\left| \int (p_n(x, x') - p(x, x')) \log(g(y, y')^{-1}) d\mu \right| < \epsilon, \quad (6.67)$$

which is the definition of continuity. The proof is finished. $\square$

Proof of Theorem 2. We prove by contradiction. Define $D_n = \{\|y\| \geq n\}$. If μ^* does not have compact support in Y then $\mu(D_n) > 0$ for all n . By lemma 17, there exists continuous $h(x)$ such that

$$\int p(x, x') \log(g(y, y')^{-1}) d\mu^* = h(x) - \frac{\int g(y, y') d\mu^*}{\int \int g(y, y') d\mu^* d\mu^*}. \quad (6.68)$$

Multiply both sides by $\mathbb{1}_{D_n(x, y)}/\mu^*(D_n)$ and integrate with respect to the other variable:

$$\begin{aligned} & \iint p(x, x') \log(g(y, y')^{-1}) \frac{\mathbb{1}_{D_n(x, y)}}{\mu^*(D_n)} d\mu^* d\mu^* \\ &= \int h(x) \frac{\mathbb{1}_{D_n(x, y)}}{\mu^*(D_n)} d\mu^* - \frac{\iint g(y, y') \frac{\mathbb{1}_{D_n(x, y)}}{\mu^*(D_n)} d\mu^* d\mu^*}{\iint g(y, y') d\mu^* d\mu^*}. \end{aligned} \quad (6.69)$$

Since h is continuous and μ_X has compact support, the first term of the RHS is bounded uniformly in n . Moreover, $g(y, y')$ is bounded so the second term is also bounded uniformly in n . It follows that the LHS is bounded uniformly in n . With this being said, because $p(x, x') > 0$ and μ_X has compact support, there exists a constant c such that $p(x, x') \geq c$ for all x, x' in the support of μ_X .

Meanwhile, since the output kernel $g(y, y') = k(\|y - y'\|)$ is integrable, it has a decaying tail and there exists $C(n)$ such that $C(n) \rightarrow \infty$ and if $\|x - x'\| \geq n/2$, $g(y, y')^{-1} = k(\|y - y'\|)^{-1} \geq \log(1 + C(n))$. We compute

$$\begin{aligned} & \frac{1}{\mu^*(D_n)} \iint p(x, x') \log(g(y, y')^{-1}) \mathbb{1}_{\|y\| \geq n} d\mu^* d\mu^* \\ & \geq \frac{1}{\mu^*(D_n)} \iint p(x, x') \log(g(y, y')^{-1}) \mathbb{1}_{\|y\| \geq n} \mathbb{1}_{\|y - y'\| \geq n/2} d\mu^* d\mu^* \\ & \geq c \frac{\log(1 + C(n))}{\mu^*(D_n)} \iint \mathbb{1}_{\|y\| \geq n} \mathbb{1}_{\|y'\| \leq n/2} d\mu^* d\mu^* \\ & = c \log(1 + C(n)) \mu^*(\|Y'\| \leq n/2). \end{aligned} \quad (6.70)$$

Clearly,

$$\lim_{n \rightarrow \infty} c \log(1 + C(n)) \mu^*(\|Y'\| \leq n/2) = \infty. \quad (6.71)$$

This contradicts with the boundedness. We must have that $\mu^*(D_n) = 0$ for n sufficiently large. That is, $\mu^*(\|Y\| \geq n) = 0$, so μ^* has compact support in Y . $\square$

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